

Advanced exponential equations

№1. Solve the following equations:

$$1. \quad \sqrt{4^x - 2^x - 3} = \sqrt{4 \cdot 2^x - 7}; \quad 2. \quad \sqrt{1 + 3^x - 9^x} = \sqrt{4 - 3 \cdot 3^x}.$$

№2. Solve the following equations:

$$1. \quad \left(\sqrt{2 + \sqrt{3}}\right)^x + \left(\sqrt{2 - \sqrt{3}}\right)^x = 4; \quad 2. \quad \left(\sqrt{4 + \sqrt{15}}\right)^x + \left(\sqrt{4 - \sqrt{15}}\right)^x = 8.$$

№3. Solve the following equations:

$$1. \quad 4^{x+\frac{1}{2}} + \frac{2}{4^x} + 14 = 9\left(2^x + \frac{1}{2^x}\right); \quad 2. \quad 9^{x+\frac{1}{2}} + \frac{3}{9^x} + 26 = 16(3^x + 3^{-x}).$$

№4. For which values of the parameter a does the equation

$$9^x - (a + 1) \cdot 3^x + 3a - 6 = 0$$

have a unique solution?

№5. For which values of the parameter a does the equation

$$25^x + 5^{x+1} - a^2 + a + 6 = 0$$

have no solutions?

№6. Find the values of the parameter a for which an equation has two different roots:

$$1. \quad 4^x - (a + 1) \cdot 2^x + 2a - 2 = 0; \quad 2. \quad (\sqrt{x} - a)(3^{2x} - 4 \cdot 3^x + 3) = 0.$$

№7. Solve the following equations:

$$1. \quad 4^x - (19 - 3x) \cdot 2^x + 34 - 6x = 0; \quad 2. \quad 9^x - (14 - x) \cdot 3^x + 33 - 3x = 0.$$

№8. Solve the following equations:

$$1. \quad 4^{\tan x} + 4^{\cot x} = 8; \quad 2. \quad 2^{\cos x} + 2^{\sin x} = 2^{1-\frac{\sqrt{2}}{2}}.$$

№9. For which values of a equations $3^x + 3^{x+3} = 3^{x+1} + 25$ and $|a - 4|2^x + a \cdot 4^x = 4$ are equivalent?

№10. Find all values of p for which the equation $(p-4)9^x + (p+1)3^x + 2p-1 = 0$ has no solutions.

№11. Find all values of a for which the equation $(a-1)3^{2x} - (2a-1)3^x - 1 = 0$ has two different solutions.

№12. Find the values of the parameter a for which an equation has a unique solution:

1. $2^{|x|} = ax^2 + a^2;$
2. $\left(\frac{1}{3}\right)^{|x|} = \left(a + \frac{1}{3}\right)x^2 + a^2;$
3. $\left(2 + \sqrt{3}\right)^x + \left(2 - \sqrt{3}\right)^x = 2(a-1)x^2 + \frac{1}{2}a^2;$
4. $\left(3 - 2\sqrt{2}\right)^x + \left(3 + 2\sqrt{2}\right)^x = (3a+1)|x| + 2a^2.$